

BANK OF CLI PRESENTATION

Thomas Nguyen >>>



PROJECT REQUIREMENTS



AUTHENTICATION

Secure user registration and login.



PERSISTENCE

View balance and transaction history.



TRANSACTIONS

ACID-compliant deposit, withdraw, and transfer.



MULTI-LAYERED

Strict separation of concerns.



LOGGING

Track application activity and errors.

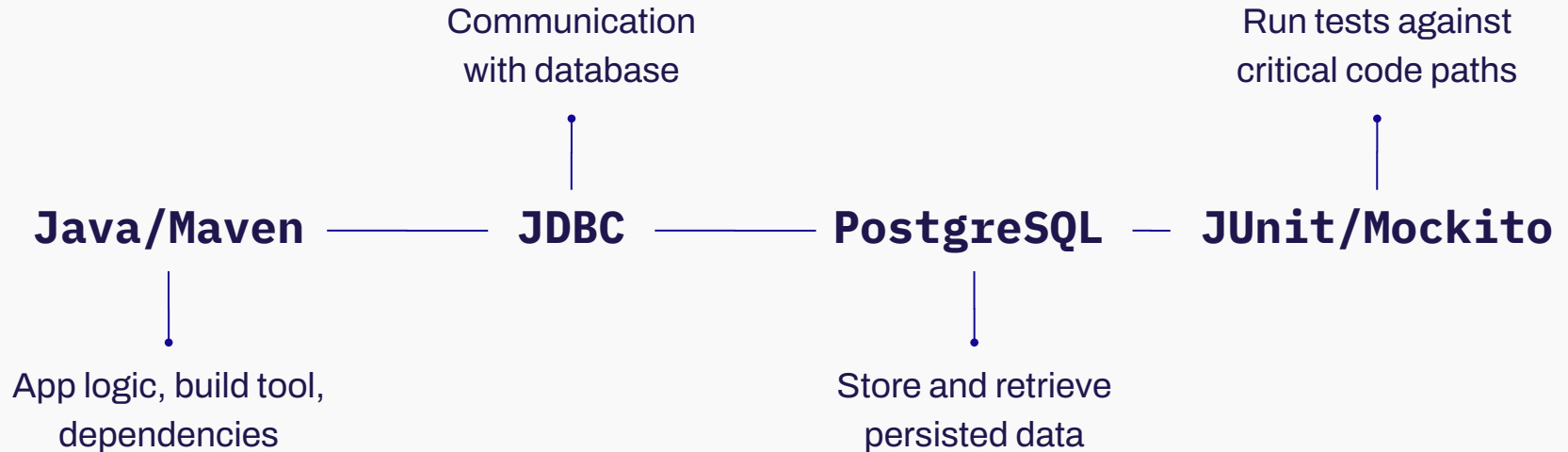


TESTING

Ensure core functionality works properly.



TECHNOLOGY STACK





PROJECT DEMO



INPUT VALIDATIONS

Please enter your account details below:

First Name: **Thomas**

Last Name: **Nguyen**

PIN: **123**

PIN must be at least 4 characters.

Enter the account ID you'd like to transfer to: **ACT-0000**

Account ID ACT-0000 could not be found.

Please choose an option below:

0. Exit
1. Login
2. Create Account

> 3

Invalid option.

Amount to withdraw: **\$9999**

Memo: **ATM**

Insufficient funds. You only have \$145.00 in your account.

Please enter your credentials below:

Account ID: **ACT-0000**

PIN: **0000**

An unexpected error occurred: Could not connect to the database.

Please enter your credentials below:

Account ID: **ACT-0000**

PIN: **0000**

Invalid credentials. Please try again.

Amount to deposit: **\$-abc.123**

Invalid format. Enter a whole amount, optionally with a decimal and two cent digits.

UNIT TESTING

JUnit

Write/run test suites against critical service methods

Mockito

Mocks, stubs, and interaction verifications

```
[INFO] Tests run: 9, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.577 s -- in io.github.tthn0.service.LedgerServiceImplTest
[INFO] Running io.github.tthn0.service.AccountServiceImplTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.014 s -- in io.github.tthn0.service.AccountServiceImplTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 11, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 8.374 s
[INFO] Finished at: 2026-09-23T15:08:27-05:00
[INFO] -----
```

LedgerService

```
@Test
void testDepositingNegativeAmountThrowsError() { ...

@Test
void testSuccessfulDeposit() { ...

@Test
void testWithdrawingNegativeAmountThrows() { ...

@Test
void testWithdrawingMoreThanYouHaveThrows() { ...

@Test
void testSuccessfulWithdraw() { ...

@Test
void testTransferringNegativeAmountThrows() { ...

@Test
void testTransferringMoreThanYouHaveThrows() { ...

@Test
void testTransferringToInvalidAccountThrows() { ...

@Test
void testSuccessfulTransfer() { ...
```

AccountService

```
@Test
void testLogin() {
    when(dao.selectAccountById(this.accountId)).thenReturn(this.account);

    Account successfulLogin = service.login(accountId, correctPin);
    Account unsuccessfulLogin = service.login(accountId, "Wrong Pin");

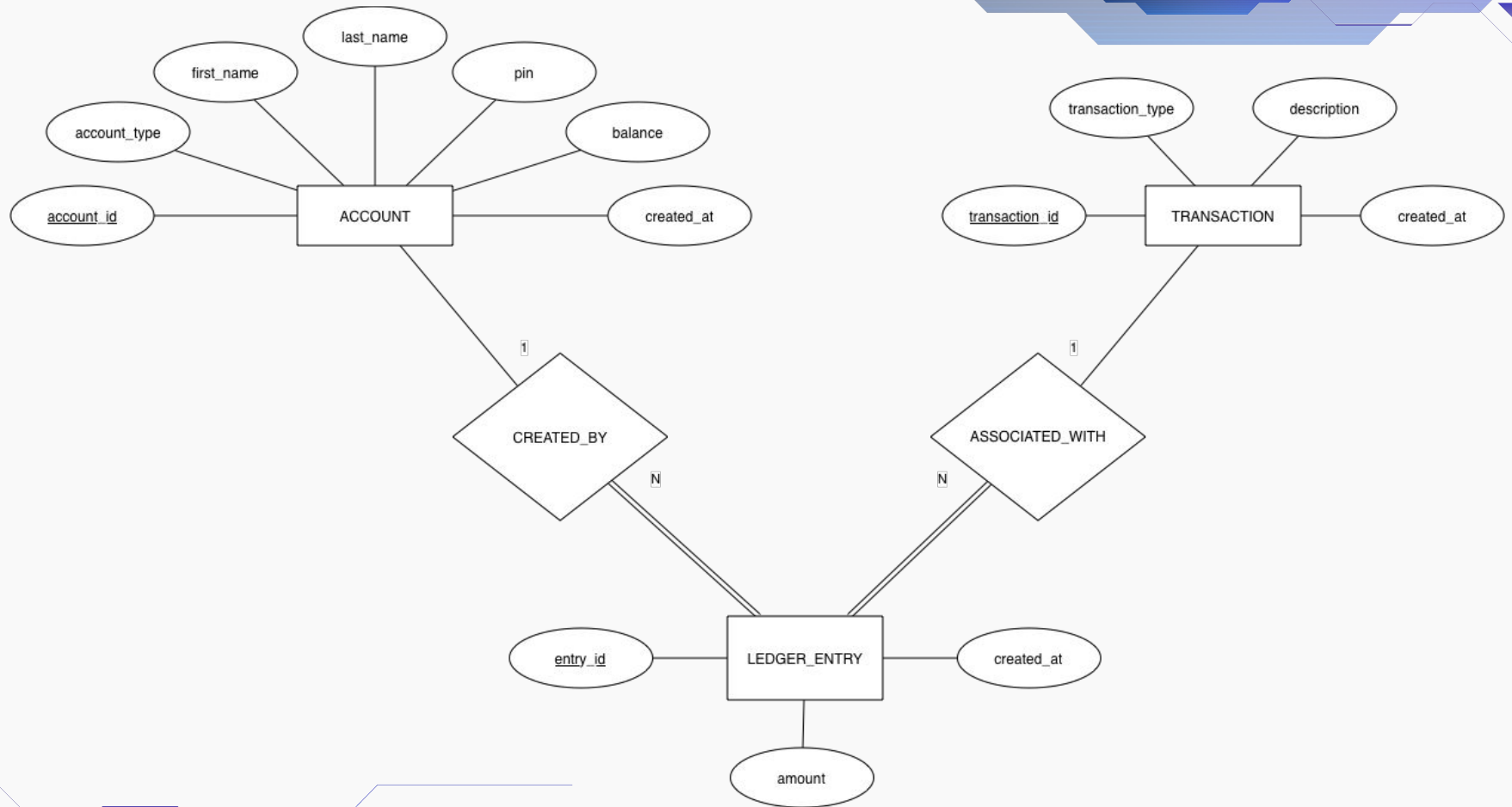
    assertNotNull(successfulLogin);
    assertNull(unsuccessfulLogin);
}

@Test
void testFindAccount() {
    when(dao.selectAccountById(this.accountId)).thenReturn(this.account);

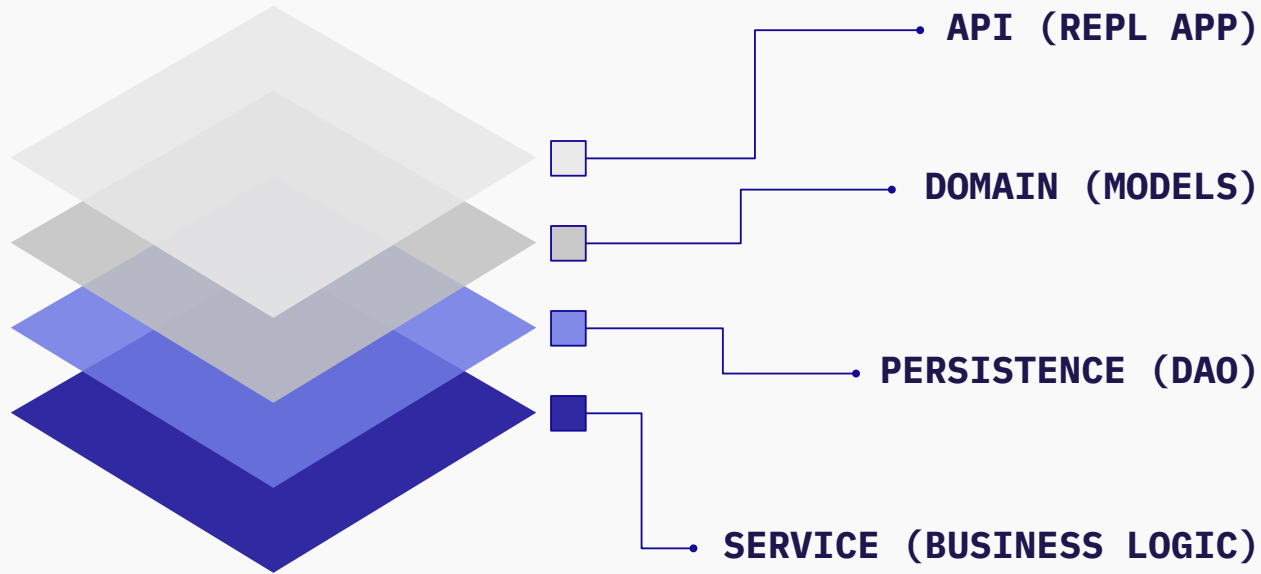
    Account existingAccount = service.findAccount(this.accountId);
    Account nonexistingAccount = service.findAccount("Non Existent");

    assertNotNull(existingAccount);
    assertNull(nonexistingAccount);
}
```





MULTI-LAYERED ARCHITECTURE



```
src
├── main
│   ├── java/io/github/tthn0
│   │   ├── app
│   │   │   ├── Main.java
│   │   │   └── Repl.java
│   │   └── domain
│   │       ├── Account.java
│   │       └── Audit.java
│   └── persistence
│       ├── AccountDao.java
│       ├── AccountDaoImpl.java
│       ├── ConnectionFactory.java
│       ├── LedgerDao.java
│       └── LedgerDaoImpl.java
└── service
    ├── AccountService.java
    ├── AccountServiceImpl.java
    ├── LedgerException.java
    ├── LedgerService.java
    └── LedgerServiceImpl.java
```

OOP + MULTI-LAYERED ARCHITECTURE

A

ABSTRACTION

DAO/service interfaces hiding implementation.

P

POLYMORPHISM

Overridden methods in service classes.

I

INHERITANCE

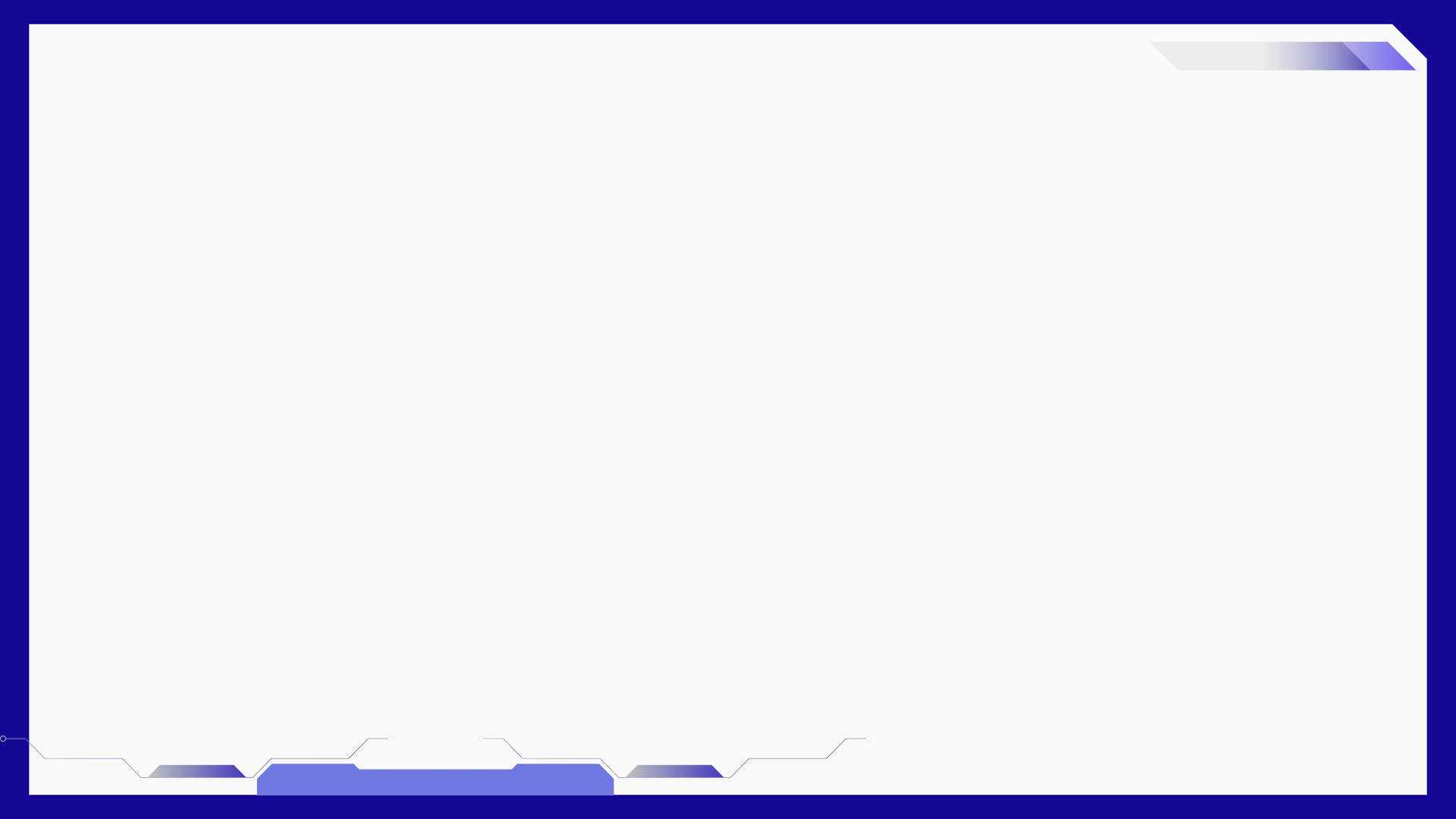
Extending base classes to create custom exceptions.

E

ENCAPSULATION

Private fields controlled by getters and setters inside models.

Q&A



DANKE !

Gibt es noch Fragen?

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